🇭🇺 Hungary — Monthly Intelligence Report v4.2

Edition 005 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · July 2026 · SSI v4.2 (refreshed June 2026)

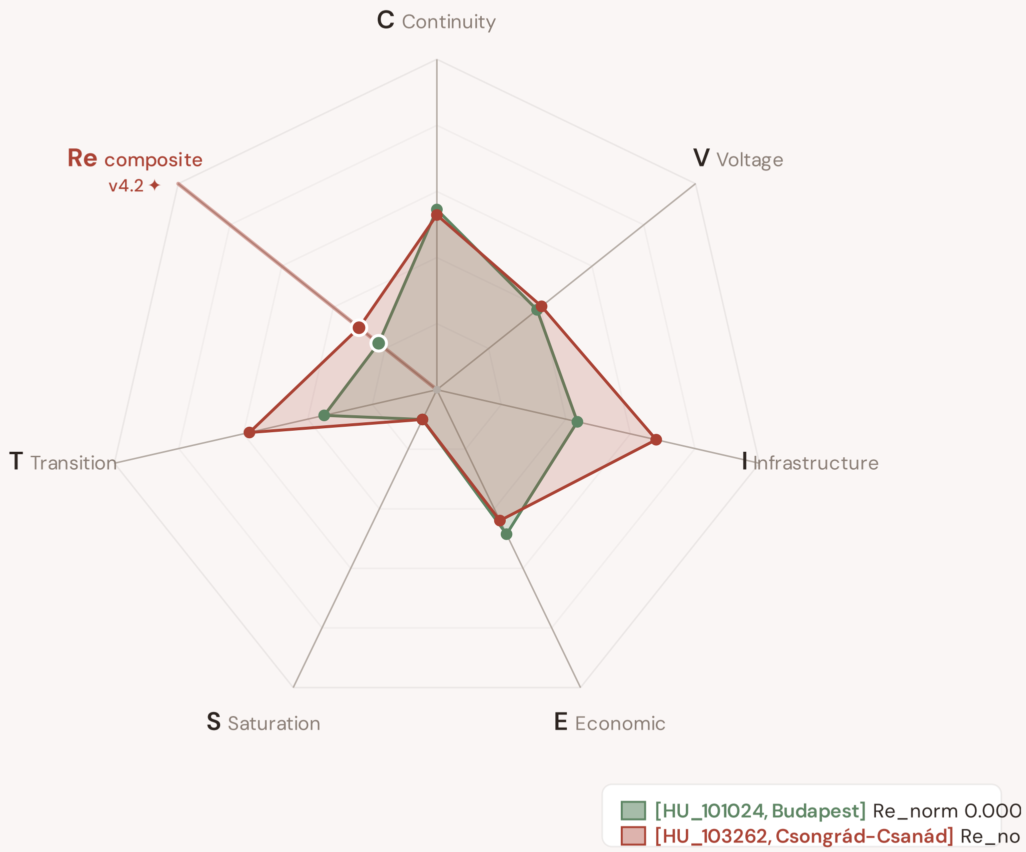
CONTENTS

SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw − 0.920) / (1.787 − 0.920), 0, 1)). Two worked examples overlaid: **[HU_101024, Budapest]** (low-exposure reference, Re_norm 0.000) · **[HU_103262, Csongrád-Csanád]** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula `Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00)`*, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared* `intelligence-sections.js` *and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).*

SUBSTATIONS SCORED

3,502

65 EHV · 307 HV · 3,130 distribution-tier

GRID LINES MAPPED

4,261

~10,635 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 3,502 substations and 4,261 grid lines spanning ~10,635 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

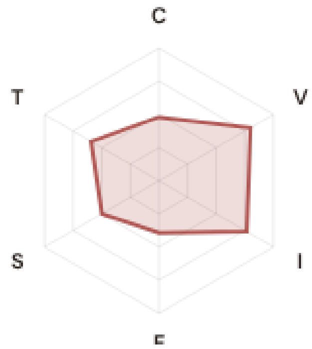
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Hungary's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

50778

Budapest, HU110 · distribution-tier

0.734	High	0.172	82th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



Re composite v4.2 7th axis

Re_raw 1.1184 · **Re_norm** 0.020

LOW EXPOSURE

COMPONENTS

- **C** Continuity: 0.479 / 0.30 (160%)
- **I** Infrastructure: 0.766 / 0.25 (306%)
- **S** Saturation: 0.500 / 0.20 (250%)
- **V** Voltage: 0.800 / 0.10 (800%)
- **E** Economic: 0.393 / 0.10 (393%)
- **T** Transition: 0.593 / 0.05 (1185%)

MODIFIERS

- **R3 Consequence:** 1.042 ↑ amplifies
- **R4 Graph Criticality:** 1.039 ↑ amplifies
- **R6a Restoration:** 0.985 ↓ dampens
- **R6b Seismic:** 1.035 ↑ amplifies
- **R7 Digital Readiness:** 1.003 → neutral

This month's spotlight — automatically selected for July 2026 — is **50778** in Budapest, HU110. With an R_final of 0.734 (High band, 82th fleet percentile), its risk profile is dominated by the T (Transition) component at 1185% of its weight ceiling, followed by V (Voltage) at 800%. Modifiers collectively shift R_base by 25.7%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

470

High/Critical + R7 < median

AVG R7 MODIFIER

0.9957

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

923

26.4% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **470 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9900). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **HU110 (151), HU120 (61), HU213 (52)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Ft5,400–10,800/kWh (HUF)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

923 substations (26.4%) High/Critical: **658** (71.3%) Avg R: **0.602** Avg E: **0.460** / 0.10

Low 7 Med 258 High 490 Crit 168

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Ft2,700–5,400/kWh (HUF)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

833 substations (23.8%) High/Critical: **535** (64.2%) Avg R: **0.579** Avg E: **0.450** / 0.10

Low 9 Med 289 High 383 Crit 152

🏪 Commercial / SME-Dense

Est. VoLL: Ft1,080–2,700/kWh (HUF)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

873 substations (24.9%) High/Critical: **566** (64.8%) Avg R: **0.575** Avg E: **0.481** / 0.10

Low 12 Med 295 High 440 Crit 126

🌾 Light / Agricultural / Rural

Est. VoLL: Ft360–1,080/kWh (HUF)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

873 substations (24.9%) High/Critical: **523** (59.9%) Avg R: **0.543** Avg E: **0.476** / 0.10

Low 9 Med 341 High 438 Crit 85

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **658 substations** combine capital-intensive economic fabric (R3 ≥ 1.04) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **HU110 (450), HU213 (90), HU120 (85)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 5.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 NUTS-3 REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Bács-Kiskun ⚠ high R	0.9918
2	Baranya ⚠ high R	0.9925
3	Jász-Nagykun-Szolnok ⚠ high R	0.9925
4	Somogy ⚠ high R	0.9926
5	Borsod-Abaúj-Zemplén ⚠ high R	0.9937
6	Fejér ⚠ high R	0.9939
7	Komárom-Esztergom ⚠ high R	0.9943
8	Hajdú-Bihar ⚠ high R	0.9945
9	Pest ⚠ high R	0.9949
10	Nógrád ⚠ high R	0.9954
11	Budapest	0.9959
12	Szabolcs-Szatmár-Bereg ⚠ high R	0.9960
13	Tolna ⚠ high R	0.9961
14	Veszprém ⚠ high R	0.9962
15	Heves ⚠ high R	0.9964

NUTS-3-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Bács-Kiskun** has the lowest average R7 (0.9918), while **Békés** leads at 1.0004 — a spread of 0.0086 across the R7 range. **0 nuts-3 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (July 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9957, median = 0.9900; 0 substations classified HIGH cyber-exposure; 470 blind-spot substations (High/Critical risk + below-median R7); 123 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **— verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	HOURLY — MONTHLY	QUARTERLY / ANNUAL
—	0	0
— variables	High-frequency feeds	Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — July 2026

OMSZ	OMSZ — Országos Meteorológiai Szolgálat	DAILY	Station	5 vars
OSM	OpenStreetMap Power Infrastructure	WEEKLY	Node/edge	3 vars
GHCN-D	NOAA GHCN-Daily (global meteorological stations)	DAILY	Station	4 vars
NASA	NASA FIRMS active-fire detection	DAILY	375 m / 1 km	2 vars
MEKH	MEKH — Magyar Energetikai és Közmű-szabályozási Hivatal	QUARTERLY	National	6 vars
KSH	KSH — Központi Statisztikai Hivatal	ANNUAL	Megye / Járás	10 vars
HEPA	Hungarian Energy Trader Association	ANNUAL	National	2 vars
OTOL	OEM — Országos Termelői Online Lakossági	ANNUAL	National	2 vars
EURO	Eurostat NUTS-3 regional database	ANNUAL	NUTS-3	5 vars
EU-SILC	EU-SILC Energy Poverty Indicators	ANNUAL	NUTS-2	3 vars
DESI	DESI Digital Economy & Society Index	ANNUAL	National	2 vars
CDS	Copernicus CDS / ERA5-Land	ANNUAL	0.1° (~11 km)	5 vars
INFORM	INFORM Risk Index (humanitarian-context)	ANNUAL	National	3 vars
IRENA	IRENA Renewable Capacity Statistics	ANNUAL	National	2 vars
IEA	IEA World Energy Outlook	ANNUAL	National	3 vars
NGFS	NGFS climate-stress scenarios	ANNUAL	National	2 vars
MBFSZ	MBFSZ — Mining and Geological Survey	STATIC	National	3 vars
GEM	GEM Global Seismic Hazard Map 2023.1	STATIC	0.05° (~5.5 km, roc...	1 var
IEEE	IEEE C57.91 / IEC 60076 (transformer thermal model)	STATIC	Asset-level	16 vars
CMIP6	CMIP6 climate-projection ensemble	STATIC	~1° grid	3 vars
PVGIS	EU JRC PVGIS solar radiation database	STATIC	~1 km	2 vars
MAVIR	MAVIR — Hungarian TSO	REAL-TIME	Bidding zone	12 vars
ENTSE	ENTSO-E Transparency Platform	HOURLY	Bidding zone	4 vars

OPEN-M	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE	MEDIAN R	HIGH + CRITICAL
3,502	0.569	2282
substations scored	P5=0.319 · P95=0.851	65.2% of fleet

HIGH CONFIDENCE

99%

3478 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 37 (1.1%) Medium 1183 (33.8%) High 1751 (50.0%) Critical 531 (15.2%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **3,502 substation scores remain unchanged** this edition. The fleet median R of 0.569 (mean 0.575) reflects a distribution skewed toward the lower bands, with 1.1% in Low and 33.8% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across P0, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	§5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	§5

V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	§5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	§5
V42-7	NEW	Re composite — bounded resilience scalar Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c - 1.00); Re_norm = clip((Re_raw - 0.920)/(1.787 - 0.920), 0, 1)	§7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	§14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	§4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 3,502 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

111

12 EHV · 99 HV

HIGH BAND

70 + 34

93.7% of corridor

CORRIDOR MEDIAN R

0.684

Fleet median: 0.569

WORST PROVINCE

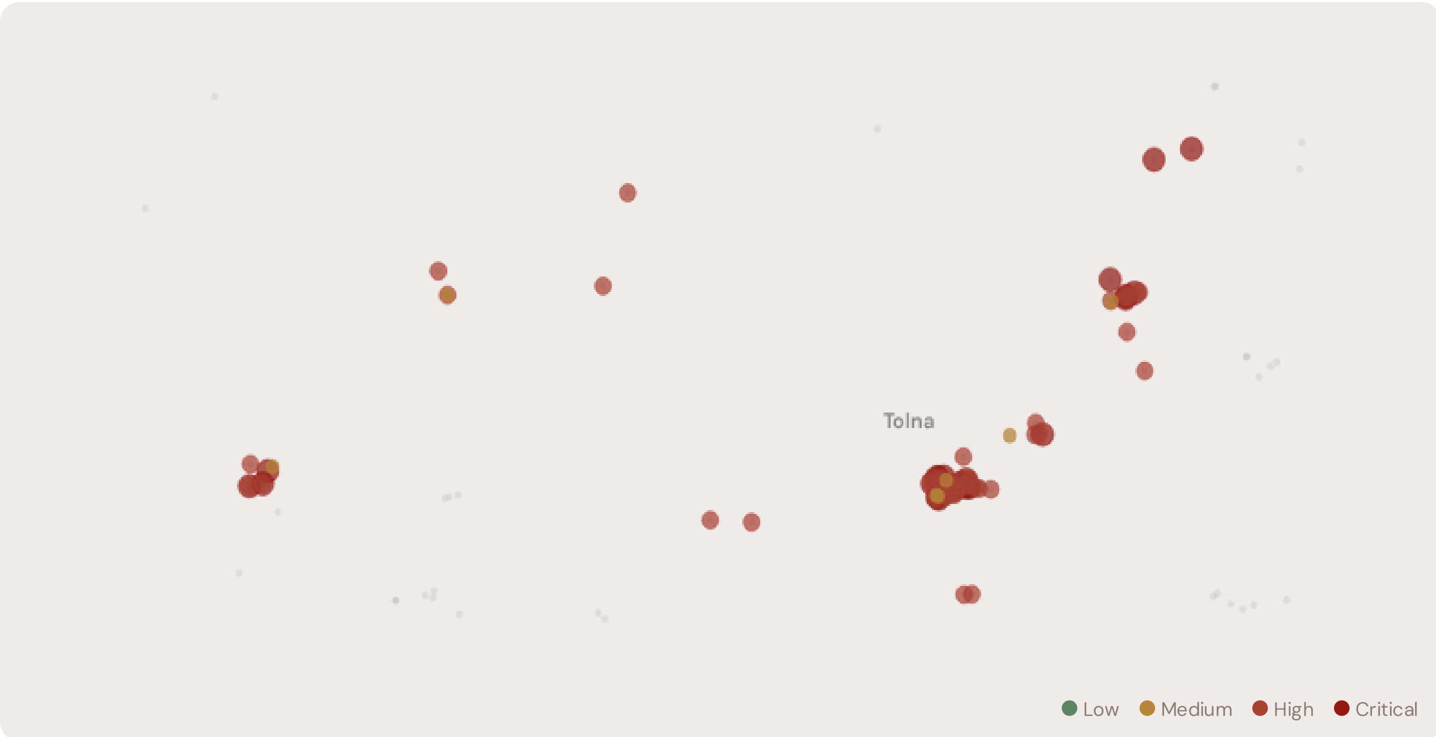
Tolna

Avg R: 0.688 · 111 substations

The Tolna megye — Paks I+II nuclear corridor hosts **111 substations** across 1 nuts-3 regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **70 substations (63.1%)** are classified High and **34** Critical, with only **0** in the Low band. 82% of corridor substations score above the national fleet median of 0.569. The corridor median R of 0.684 reflects the local risk profile. Tolna megye (HU233) is Hungary's single-site nuclear concentration zone. All four Paks I VVER-440/V-213 units operate at a single site on the Danube's right bank, ~100 km south of Budapest —

Units 1-4 commissioned 1982-1987, ~500 MWe each post-uprate, combined ~1,916 MWe generating ~45-50% of Hungarian electricity. Paks II Units 5+6 (VVER-1200/V-491, ~1,200 MWe each) are co-located ~500 m from Paks I; Unit 5 First Concrete was poured 5 February 2026 by Rosatom — the first new-NPP first-concrete in any EU country since Hinkley Point C Unit 2 (Dec 2019), and the first Russian-built NPP in an EU member state. Target operation ~2031-2032 for Unit 5, ~2032-2033 for Unit 6. US OFAC sanctions on the project were lifted in Dec 2025. The Light-Rural R3 tier (1.06) reflects Tolna's predominantly agricultural character beyond the nuclear site, but the single-site concentration drives elevated R3 + R6_seismic + R6c_flood (Danube right-bank) modifier sensitivity at the corridor level. OAH safety oversight, ENTSO-E reserve obligations and MAVIR balancing cascade across the E.ON Dél-dunántúli DSO simultaneously in any Paks trip scenario. When Paks II Units 5+6 come online, HU nuclear share could approach 70%+ — among the world's highest.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE V4.2	ALERT
HU_102394	Tolna	0.980	Critical	1.000	0.533	1.000	1.000	0.500	1.000	C	
HU_102414	Tolna	0.971	Critical	1.000	0.514	0.820	0.784	0.500	0.900		
HU_102435	Tolna	0.956	Critical	0.798	1.000	0.937	0.523	0.500	0.485	V	
Paks DÉDÁSZ állomás	Tolna	0.937	Critical	1.000	0.856	0.692	1.000	0.500	0.623	C	
14615	Tolna	0.932	Critical	0.903	0.798	0.949	0.576	0.500	0.570		
Szekszárd alállomás	Tolna	0.871	Critical	1.000	0.688	1.000	0.303	0.500	0.909		
HU_102663	Tolna	0.867	Critical	1.000	0.566	0.875	0.620	0.500	0.533		
HU_102450	Tolna	0.864	Critical	0.260	1.000	1.000	0.880	0.500	1.000	V	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE V4.2	ALERT
HU_102526	Tolna	0.860	Critical	1.000	0.556	0.568	0.692	0.500	0.329	C	
19908	Tolna	0.859	Critical	0.963	0.346	0.713	0.857	0.500	1.000		

… 101 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — TOLNA MEGYE — PAKS I+II NUCLEAR CORRIDOR VS FLEET

C — Continuity 0.603 vs 0.475 (+27%)

I — Infrastructure 0.642 vs 0.465 (+38%)

S — Saturation 0.500 vs 0.500 (0%)

V — Voltage 0.551 vs 0.467 (+18%)

E — Economic 0.689 vs 0.467 (+48%)

T — Transition 0.651 vs 0.475 (+37%)

MODIFIER AVERAGES — TOLNA MEGYE — PAKS I+II NUCLEAR CORRIDOR VS FLEET

R3 Consequence	1.000	fleet 1.000	→
R4 Graph Crit.	0.979	fleet 0.981	↓
R6a Restoration	1.013	fleet 1.021	↑
R6b Seismic	1.020	fleet 1.034	↑
R7 Digital Read.	0.996	fleet 0.996	→

The dominant risk driver across the Tolna megye — Paks I+II nuclear corridor corridor is **T (Transition)**, averaging 0.651 against a fleet average of 0.475 — occupying 1302% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.689 vs fleet 0.467 (689% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.000 (fleet: 1.000), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.020, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

NUTS-3 RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Tolna 111 subs · avg R = 0.688 · H:70 C:34 M:7

The nuts-3 regions-level breakdown reveals significant intra-regional variation. **Tolna** leads with a mean R of 0.688 across 111 substations, while **Tolna** scores 0.688 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Tolna megye — Paks I+II nuclear corridor is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

66 Tolna megye — Paks I+II nuclear corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Tolna sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Hungary's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.

🇩🇪 Germany	11.7
🇨🇭 Switzerland	~15
🇳🇱 Netherlands	23.9
🇫🇷 France	~36
🇬🇧 UK	~40
🇮🇹 Italy	~43
🇪🇸 Spain	~48
🇭🇺 Hungary	~57
🇵🇱 Poland	~95
🇷🇴 Romania	~290

Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

This month's key insight: The Tolna megye — Paks I+II nuclear corridor corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.475. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Counties Risk Ranking — All 20 Hungarian Counties

Where does this corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1	HU332	21 subs · avg R = 0.793 · H:6 C:15
#2	HU323	33 subs · avg R = 0.773 · H:15 C:17
#3	HU313	15 subs · avg R = 0.758 · H:8 C:7
#4	HU311	96 subs · avg R = 0.750 · H:44 C:48
#5	HU231	66 subs · avg R = 0.728 · H:32 C:31
#6	HU312	38 subs · avg R = 0.717 · H:21 C:16
#7	HU232	58 subs · avg R = 0.704 · H:31 C:23
#8	HU322	30 subs · avg R = 0.701 · H:19 C:9

#9 HU331 — 52 subs · avg R = 0.691 · H:33 C:17

#10 HU333 — 105 subs · avg R = 0.689 · H:65 C:31

#11 HU233 — 111 subs · avg R = 0.688 · H:70 C:34

#12 HU321 — 114 subs · avg R = 0.677 · H:69 C:34

#13 HU223 — 39 subs · avg R = 0.655 · H:23 C:9

#14 HU213 — 349 subs · avg R = 0.650 · H:206 C:91

#15 HU222 — 144 subs · avg R = 0.646 · H:87 C:35

#16 HU212 — 51 subs · avg R = 0.628 · H:32 C:9

#17 HU120 — 346 subs · avg R = 0.610 · H:193 C:64

#18 HU221 — 44 subs · avg R = 0.595 · H:23 C:7

#19 HU211 — 85 subs · avg R = 0.590 · H:50 C:12

#20 HU110 — 1705 subs · avg R = 0.484 · H:724 C:22

The Tolna megye — Paks I+II nuclear corridor ranks **#0 of 20 nuts-3 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are HU332, HU323, HU313, while the three most resilient are HU110, HU211, HU221. 19 of 20 score above the fleet average of 0.575. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

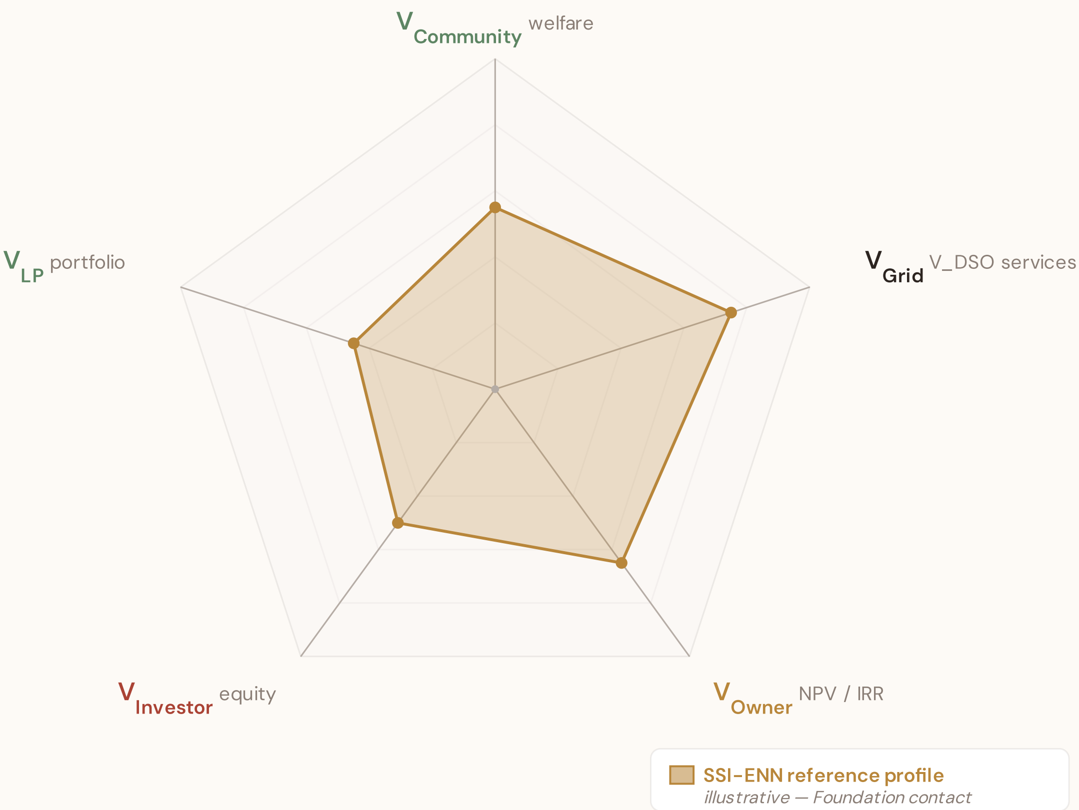
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **$V_{Community}$** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **$V_{Investor}$** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_DSO captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/TSO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

→ Quantile regression (pinball loss) + conformal prediction

→ Calibrated against SSI Monte Carlo ground truth

→ Per-country, per-voltage, per-band calibration verification

→ Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 3,502 substations with complete feature vectors. Average CI_width of 0.140 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 37 Low, 1183 Medium, 1751 High, 531 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 2 Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill
W1 · W2 · W3 · W4 · W8
Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill + chevron corner marker
W9 — Supply Security
Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill
W5 — Digital Inclusion
Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded
W6 · W7 · W10
Commercial deliverables — please contact the Foundation.

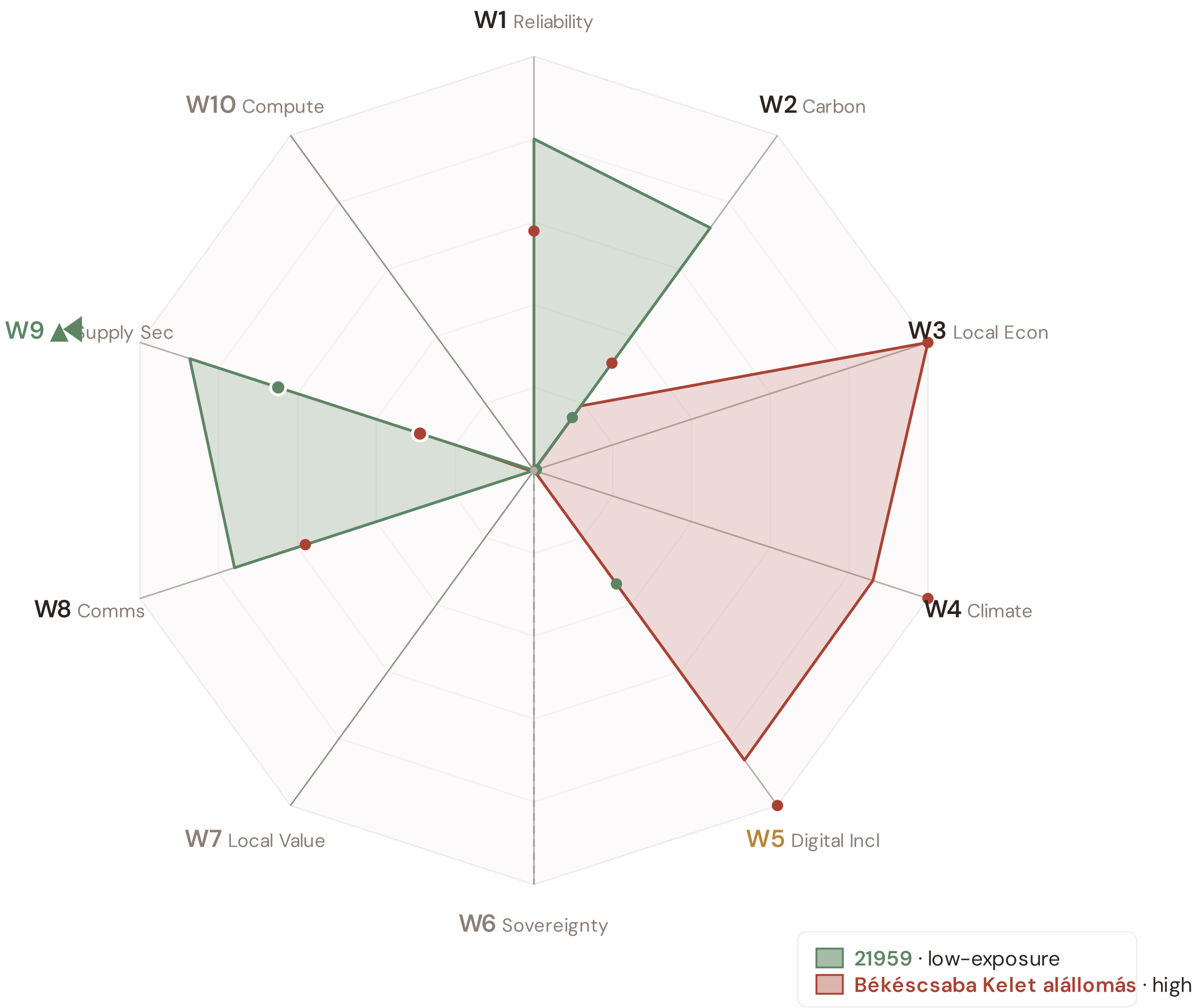
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across Hungary, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to **[0, 1]** via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (`SSI_v4_0 Italy/.../scoring-it/normalise.py`); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10-d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **[HU_101024, Budapest]** (low-exposure reference, Re_norm 0.000) and **[HU_103262, Csongrád-Csanád]** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



21959 polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Békéscsaba Kelet alállomás polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	**Published**	C × E2_local × population (SAIDI + + -POV)
W2	Carbon	**Published**	T1 × zonal MEF × (V · deg / 100) (+ EEA)
W3	Local Economic	**Published**	norm(R10_just) — EU-SILC + OIPE energy-poverty
W4	Climate Resilience	**Published**	(R6d − 1) + (R6e − 1) + (R6c − 1) + ½(R9 − 1) — v4.2 resilience modifier family
W5	Digital Inclusion	**Baseline-only**	norm(R8_adapt) — bulk SM + + DESI
W6	Sovereignty	**Commercial**	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	**Commercial**	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	**Published**	C × p_crit_10yr × population (Markov 10-yr critical probability)
W9 ▲	Supply Security	**Published-baseline ▲**	0.60 × BC_pct + 0.30 × bridge + 0.10 × (1 − deg / 20) — OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	**Commercial**	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — 21959 (low-exposure reference) and Békéscsaba Kelet alállomás (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6

§2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

The Capital Load & 2013 Danube Flood

Deep-dive into Budapest's grid risk profile — substation map, component analysis, nuts-3 regions breakdown. European Context Card: MOL refinery Százhalombatta + Suzuki Esztergom commuter + 6 Jun 2013 Vigadó peak; dyke modernisation program.

NEXT DATA REFRESH

Q4 2026 — 0 Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **KSH — Központi Statisztikai Hivatal** (10 variables → R3 GDP per capita, demographics, energy poverty).

FLEET SPOTLIGHT

611 Bridge Substations — Single Points of Failure

611 substations are topological bridges — their failure disconnects part of the network. Of these, **406** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 02 will be published on the second Thursday of August 2026. Deep-dive region: **Budapest**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Hungarian utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.